

CONTENT

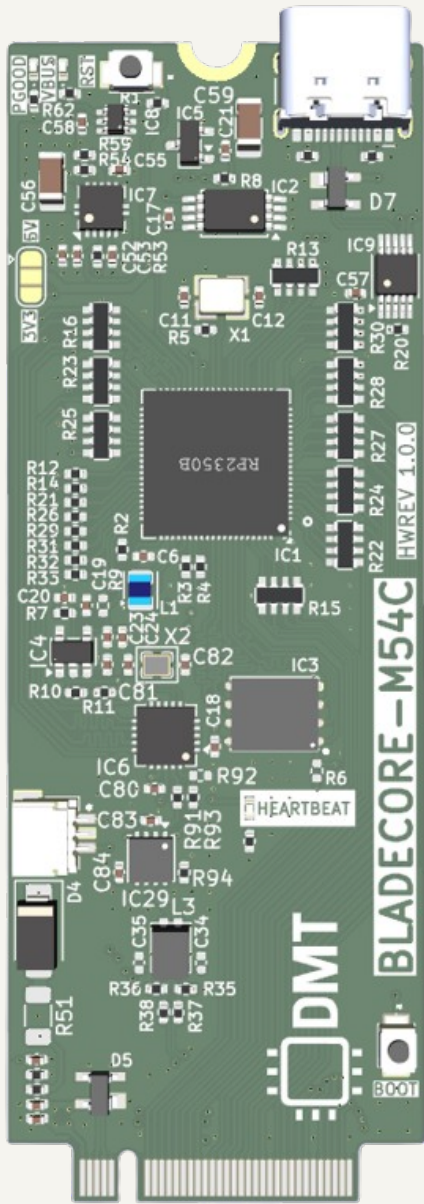
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COMMENT GUIDELINES

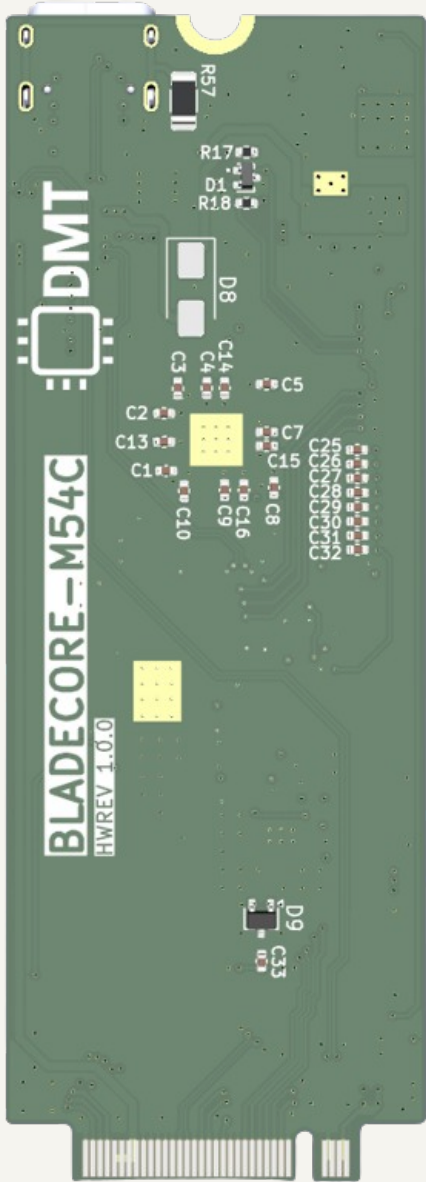
General comments are black, 50 mil size
Design notes and guidelines are blue, 50 mil size
Layout instructions are red, 50 mil size

PCB PREVIEW

TOP VIEW



BOTTOM VIEW

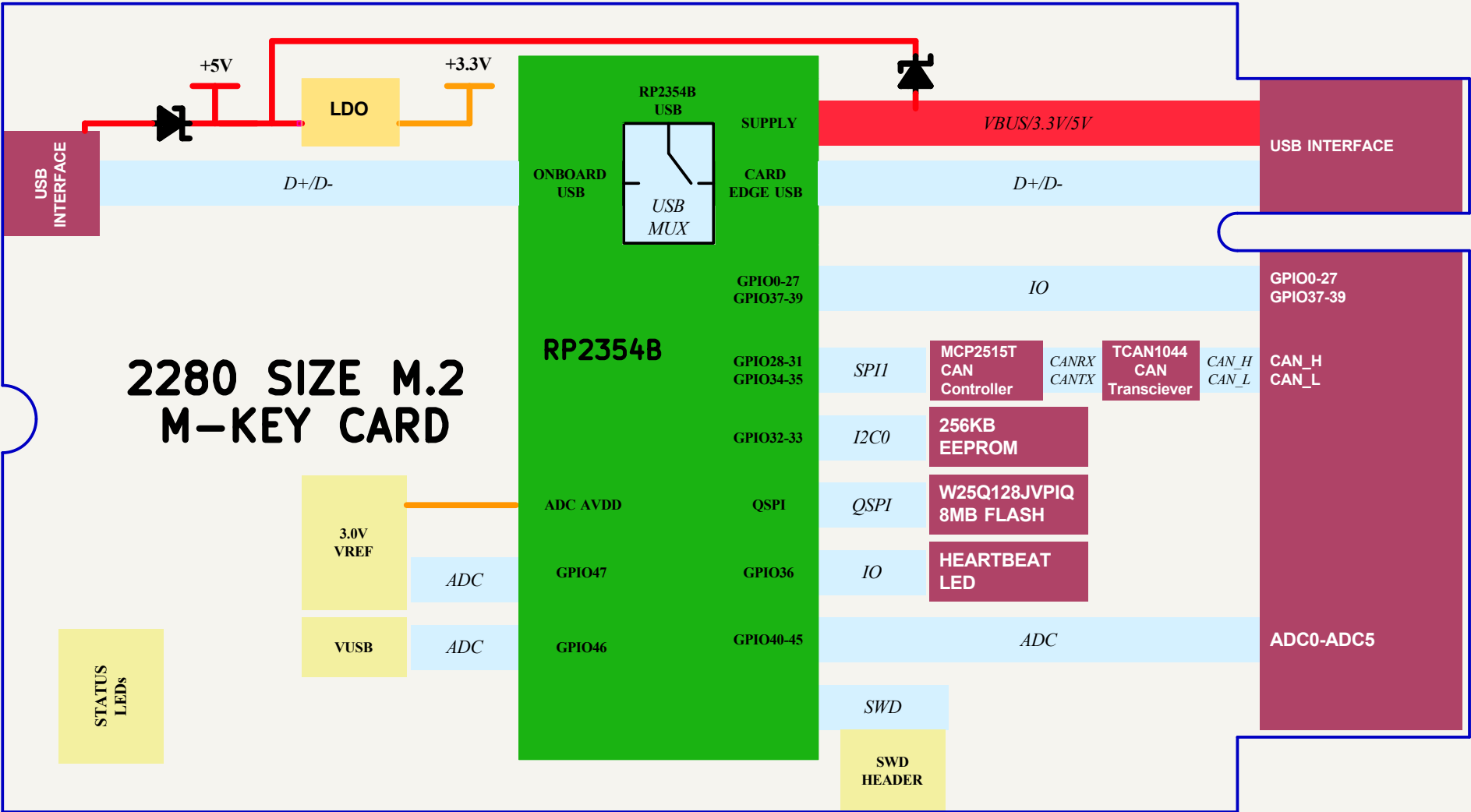


NOTES

Not fitted components are marked as

 Date: 2026-01-31 Sheet Title: BladeCore-M54C	Board Name: BladeCore File Name: BladeCore-M54C.kicad_sch		Project Name: M54 Revision: 1.0.0	
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 1 of 9

[2] Block Diagram

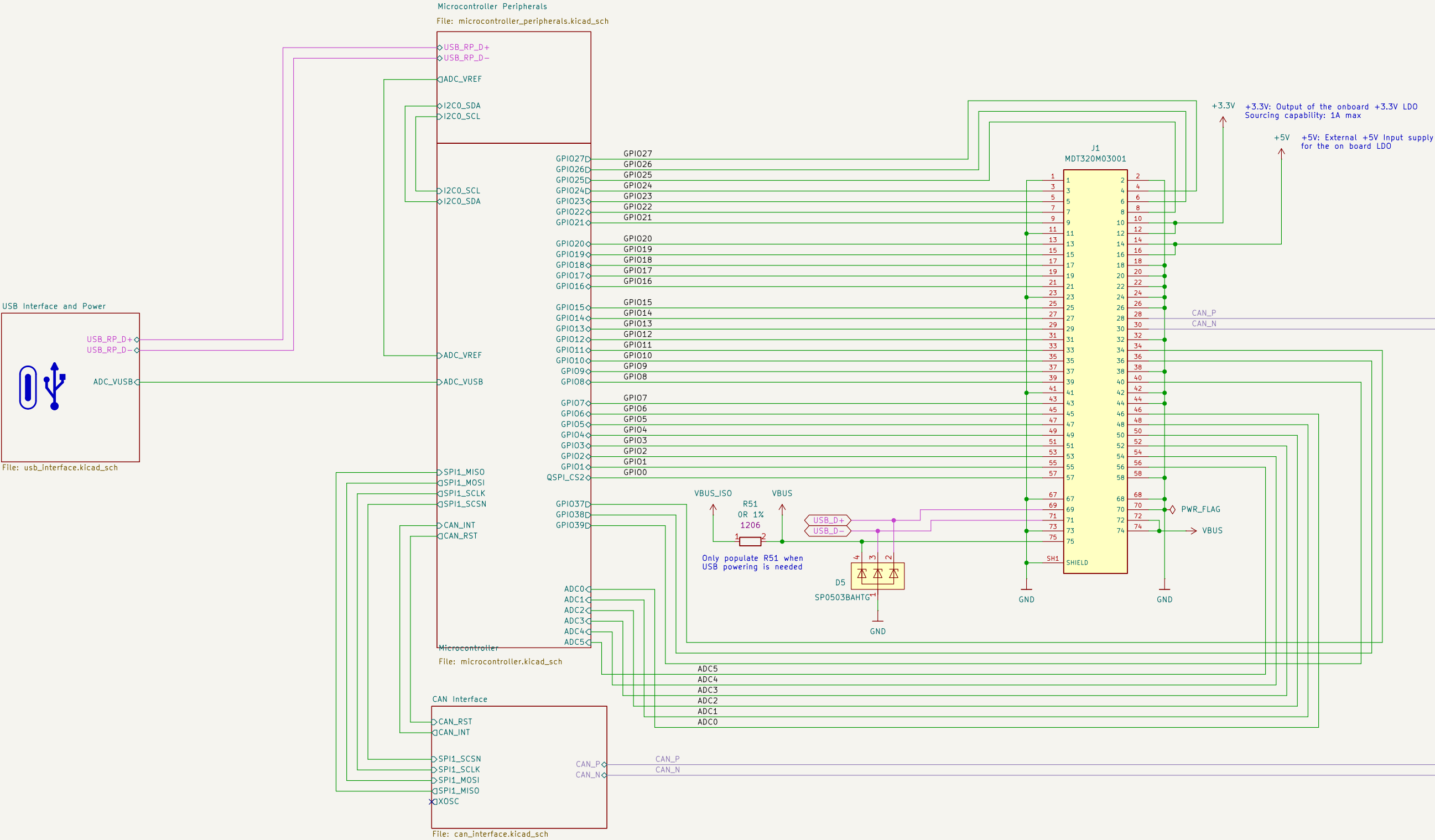


Electrical specifications:

Input voltage:	3.3V - 5.5 V
Absolut maximum current consumption	~200 mA
Typical current consumption	60mA
IO voltage	3.3V
Onboard LDO output capacity	1.5A
LDO Dropout	60mV
EEPROM	256 kbit
External Flash	8MB

 Date: 2026-01-31 Sheet Title: Block Diagram	Board Name: BladeCore		Project Name: M54	
	File Name: Block Diagram.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 2 of 9

[3] Project Architecture

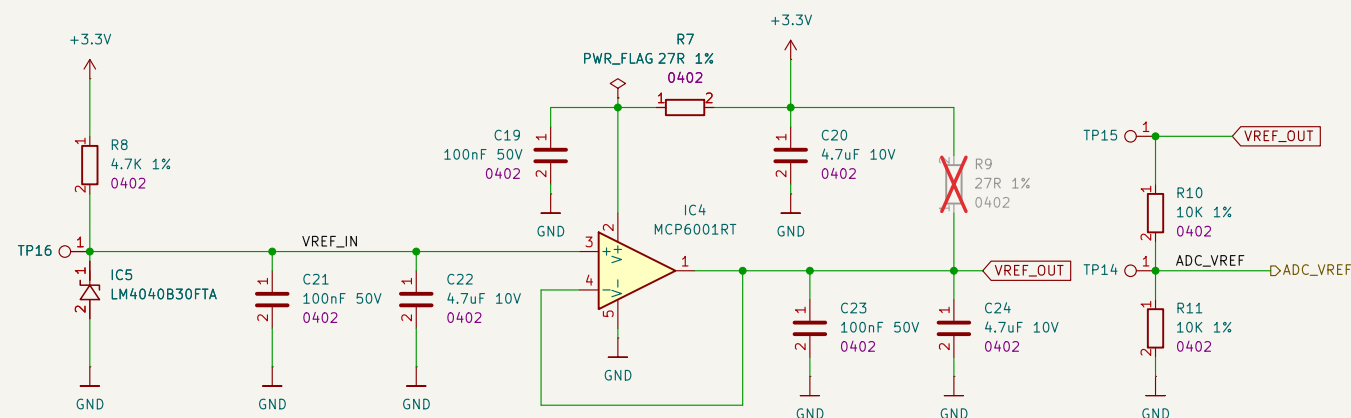


 Date: 2026-01-31 Sheet Title: Project Architecture	Board Name: BladeCore		Project Name: M54	
	File Name: Project Architecture.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 3 of 9

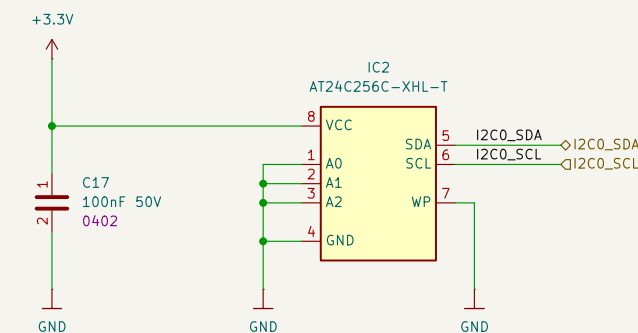
RP2354B MICROCONTROLLER



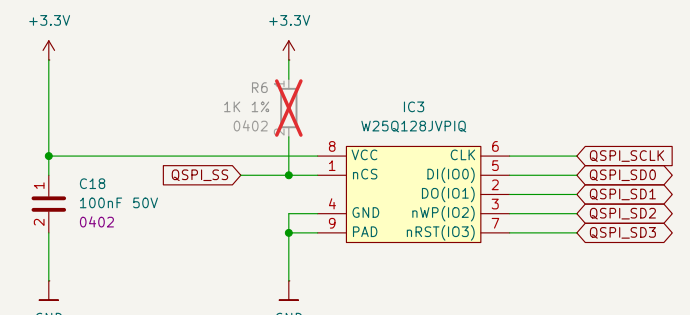
3.0V ADC VOLTAGE REFERENCE



EXTERNAL EEPROM



EXTERNAL FLASH



Sheet Title:

Microcontroller Peripherals

Board Name:
BladeCore

File Name:
microcontroller_peripherals.kicad_sch

Company:
DvidMakesThings

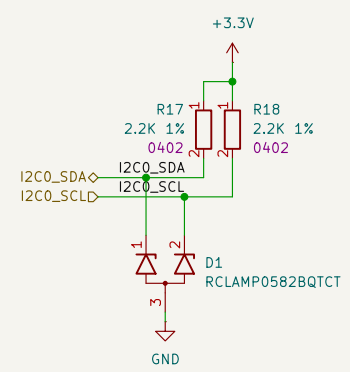
Project Name:
M54

Revision:	Variant:
1.0.0	

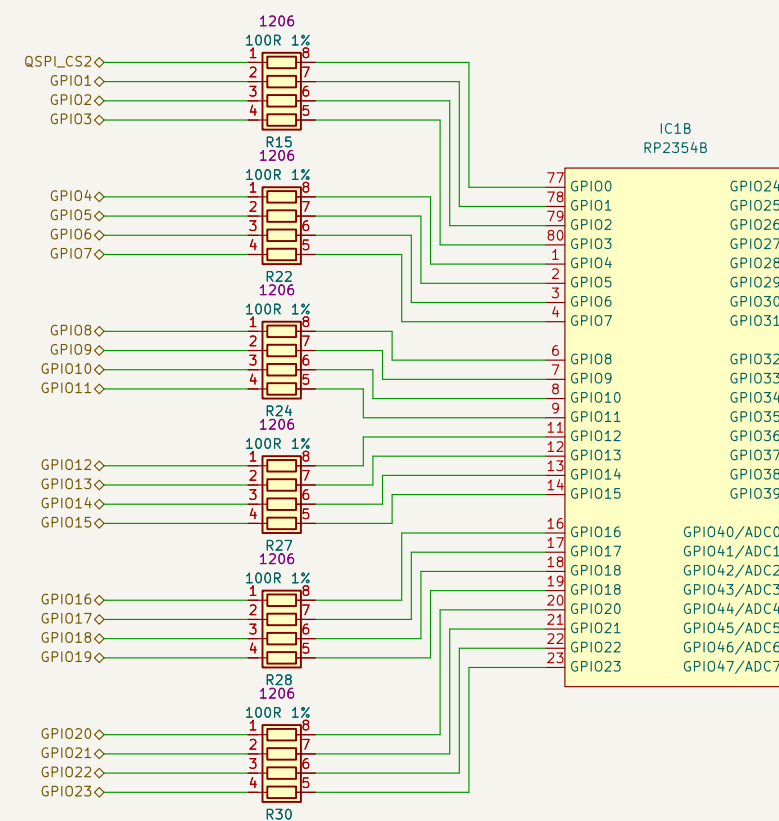
Size:	Sheet:
A3	4 of 9

[5] Microcontroller

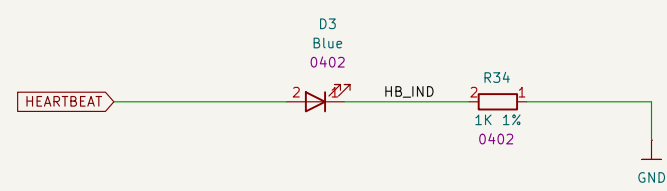
I2C PULLUP



RP2354B GPIO ASSIGNMENT



HEARTBEAT LED

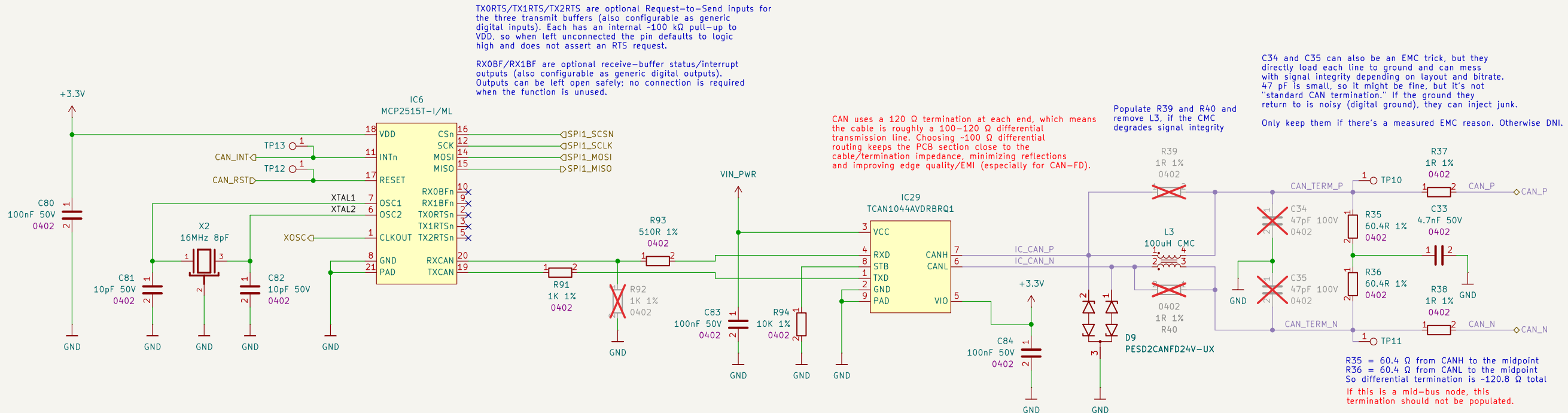


 Date: 2026-01-31 Sheet Title: Microcontroller	Board Name: BladeCore		Project Name: M54	
	File Name: microcontroller.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 5 of 9

[6] CAN Interface

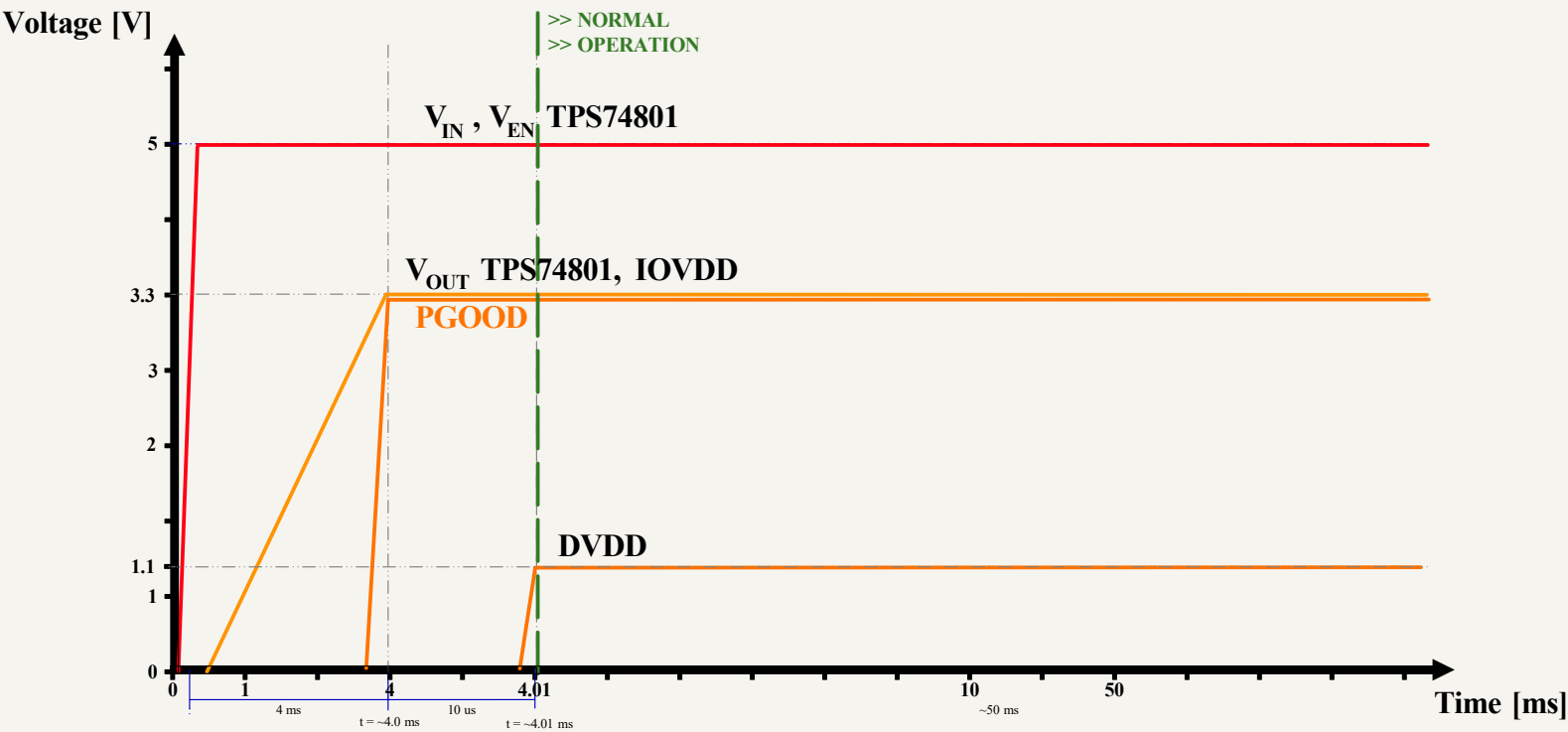
MCP2515 CAN CONTROLLER

TCAN1044 CAN TRANSCIEVER



 Date: 2026-01-31	Board Name: BladeCore		Project Name: M54	
	File Name: can_interface.kicad_sch		Revision: 1.0.0	Variant:
	Sheet Title: CAN Interface	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3
				Sheet: 6 of 9

[8] Power - Sequencing



#	Stage	Δt	Total elapsed
1	VIN applied (5.0V)	0	0.000 ms
2	VEN high (TPS74801 EN)	0	0.000 ms
3	VOUT = 3.3V (soft-start)	4.000 ms	4.000 ms
4	PGOOD valid	0	4.000 ms
5	RP2354B 3.3V present	0	4.000 ms
6	RP2354B 1.1V valid (POR)	0.010 ms	4.010 ms



Date: 2026-01-31

Sheet Title:
Power - Sequencing

Board Name:
BladeCore

File Name:
Power - Sequencing.kicad_sch

Company:
DvidMakesThings

Designer:
David Sipos
Reviewer:

Project Name:
M54

Revision:
1.0.0

Size:
A3

Sheet:
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[9] Revision History

DATE	REVISION	RESPONSIBLE	CHANGE
31.01.2026	1.0.0	DMT	INITIAL CREATION

 DMT Date: 2026-01-31 Sheet Title: Revision History	Board Name: BladeCore		Project Name: M54	
	File Name: Revision History.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 9 of 9